

DAVAO CITY MUNICIPAL ORDINANCES

Clean text. Full corpus.

Two comparable model paths.

Lexical TF-IDF + c-TF-IDF vs Legal-BERT + BERTopic

2,118 source documents · 7,832 modeled units

2016–2025 · Corpuz · Lamsin · Nuril

USER-VERIFIED TXT SOURCE / AUDITED PREPARATION / EXPERT MODEL VALIDATION PENDING

What changed in this rerun

Only data/processed/clean_text/*.txt supplies model text.

The full eligible corpus replaces the previous 200-unit sample.

Content hashes remove duplicate copies without deleting sources.

Human year overrides take priority; every decision is auditable.

Specified boilerplate sections and signature blocks are excluded.

GPU batching and checkpointed embeddings support reproducible full runs.

Both models use identical units, years, vocabulary and evaluation inputs.

Exactly which data reached the models?

Input: 2,283 TXT files → 2,133 distinct contents.

150 duplicate copies removed from modeling, not from disk.

13 content groups excluded by existing human out-of-scope year corrections.

2 content groups had no usable text after deterministic extraction.

Modeled: 2,118 source documents / 2,073 distinct normalized ordinance numbers.

7,832 units; all eligible years included with their natural corpus sizes.

Verification is based on the researcher's confirmation of the cleaned source—not a new independent PDF audit.

Year-by-year corpus coverage

Actual corpus coverage · no year-balanced sampling

Study year	Modeled text units	Source documents
2016	301	174
2017	463	204
2018	679	249
2019	984	266
2020	713	165
2021	1088	233
2022	1204	279
2023	950	218
2024	1300	298
2025	150	32

Source verification and extraction are different

7,235 automatically detected numbered sections.

597 whole-document fallback units for missing or repeated section headings.

623 effectivity/separability/repealing-clause units removed.

Explicit signature blocks are trimmed; source TXT files remain unchanged.

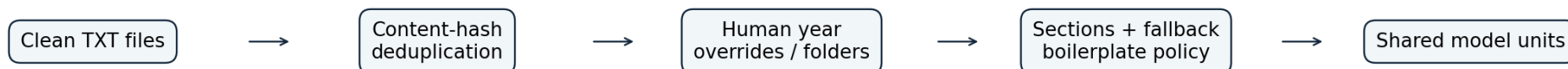
45 additional text versions share ordinance numbers and are flagged, not silently merged.

28 extracted enactment-year candidates disagree with assigned study years. Human overrides / user-verified folders remain authoritative.

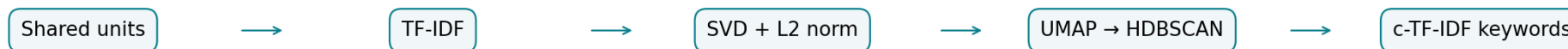
These flags and mixed unit lengths remain interpretation limitations.

A clean-text-only architecture

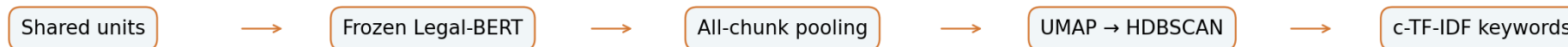
Shared preparation



Model A · Lexical



Model B · Legal-BERT



c-TF-IDF requires classes: lexical TF-IDF/SVD supplies initial assignments; both paths apply c-TF-IDF after clustering.

Legal-BERT: what actually ran

Encoder: nlpaueb/legal-bert-base-uncased; pinned revision.

11,546 non-overlapping chunks, maximum 512 tokens including specials.

Content-token weighted pooling excludes padding and special tokens; every chunk is included.

L2-normalized 768-dimensional vectors retain source file / unit / year associations.

Device: cuda · GPU inference uses unchanged pretrained weights.

No classifier training, learning rate or fine-tuning epochs are claimed. Encoder suitability still requires empirical and legal-expert assessment.

Actual tuning parameters

Executed parameter configurations

Trial	UMAP neighbors	Min distance	Cluster min size	Min samples
initial	15	0.0	15	5
conservative	30	0.1	30	10

Shared: seed 42 | UMAP 5 dimensions | top 10 unigram words | 3 seed runs | no encoder fine-tuning

Rendered configuration evidence—not a training-interface screenshot. Three stability runs means one primary fit plus two extra seeds.

Initial full-corpus metrics

Initial full-corpus configuration

Metric	Model A · Lexical	Model B · Legal-BERT
C_v coherence	0.788	0.781
Top-10 diversity	0.469	0.492
Assigned unit share	0.905	0.831
Outlier share	0.095	0.169
Seed stability (ARI)	0.530	0.574
Topics (excluding outliers)	202	159

Identical 7,832 text units for both paths; quality scores are intrinsic and in-sample.

These are in-sample intrinsic scores, not accuracy/F1 or proof of legal validity.

Read the comparison as trade-offs

Coherence C_v: lexical 0.788 vs Legal-BERT 0.781.

Keyword diversity: lexical 0.469 vs Legal-BERT 0.492.

Coverage: lexical 90.5% vs Legal-BERT 83.1%.

Seed stability ARI: lexical 0.530 vs Legal-BERT 0.574.

The stronger model can differ by criterion. More coverage or fewer topics alone does not establish better policy themes.

Do not attribute differences from the old sample solely to the text-input change: corpus size, extraction and parameter settings also changed.

Sensitivity to a second configuration

Measured parameter sensitivity · full eligible corpus

Trial	Model	Topics	C_v	Diversity	Coverage	Seed ARI
initial	lexical	202	0.788	0.469	0.905	0.530
initial	semantic	159	0.781	0.492	0.831	0.574
conservative	lexical	107	0.796	0.569	0.867	0.611
conservative	semantic	64	0.817	0.512	0.914	0.856

Joint configuration changes, not isolated ablations. No held-out winner is selected.

Both configurations are evaluated on the same units. This is not held-out tuning or a final model selection.

A real BERT output snippet

```
ENCODER  nlpaueb/legal-bert-base-uncased
DEVICE   cuda | 2.8.0+cu126
SOURCE   2016/Ordinance No. 001-16.txt / section_1
OUTPUT   7,832 units x 768 dimensions | 11,546 chunks
VECTOR   [-0.02532  0.02328  0.02861  0.01416  0.0568   0.0014   0.01647  0.03044] ...
NORM     1.000000 | TOPIC 10
```

```
TITLE.- This Ordinance shall be known as uAI{ oRDINANCE GRANTING THE RTQUEST OF THE DAVAO CITY SMALL
AND MEDIUM EIYTERPRISES DIIIELOPMEITT COUNCIL FOR EXEMPTION FROM THE PAY'MENT OF SPECIAL PERMIT
TEF,S OF ALL THE PARTICIPANTS, IN CONNECTIOIV WITH THE PINOY SME EXPO OIIT JULY 8.14, 2016, A
```

Representative outputs retain TXT provenance

Model A · Lexical

T0: authorizing, memorandum, agreement

2016/Ordinance No. 0128-16.txt

section_1 / study year 2016

TITLE.- This Ordinance shall be known as "AN
ORDINANCE GRANTING LEGISLATIVE AUTHORITY TO
THE CITY MAYOR TO ENTER INTO AND SIGN, FOR AND
IN BEHALF OF THE CITY GOVERNMENT OF DAVAO THE
MEMORANDUM OF AGREEMENT TO BE ENTERED INTO
BY AND BETWEEN THE CITY GOVERNMENT OF DAVAO
AND THE DEPARTMENT OF INFORMATION

Machine-generated keyword sets are not expert-approved policy labels.

Model B · Legal-BERT

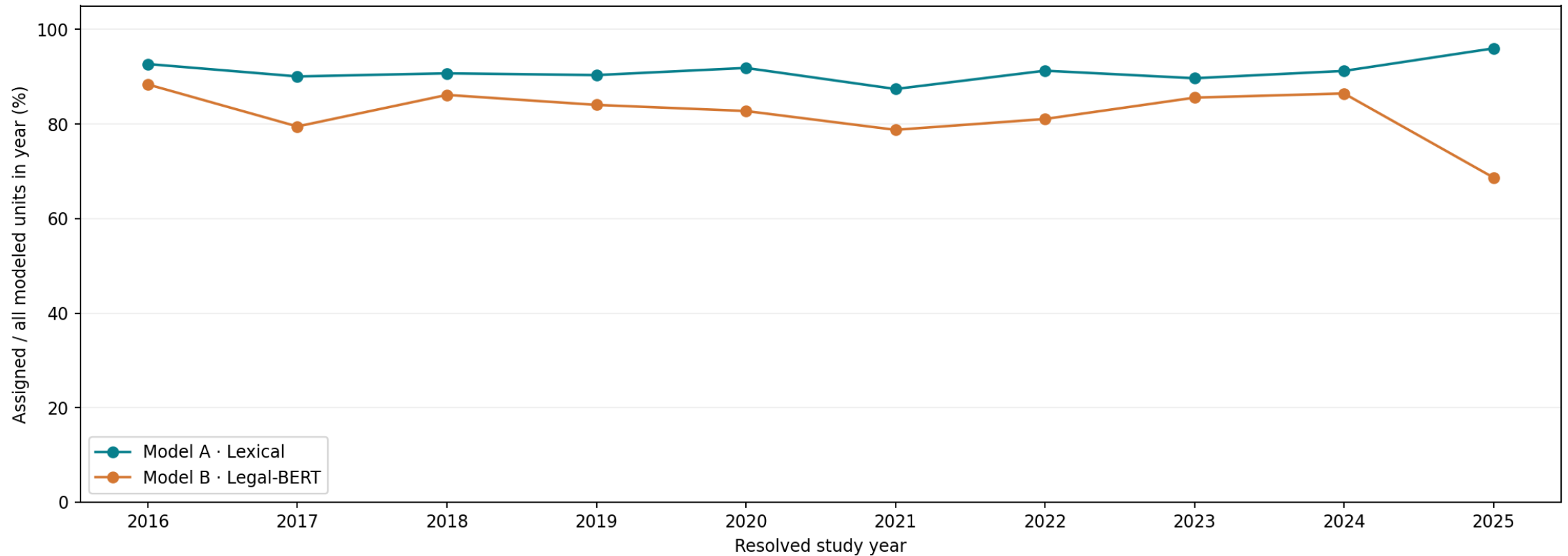
T0: street, closure, temporary

2016/Ordinance No. 100-16.txt

section_1 / study year 2016

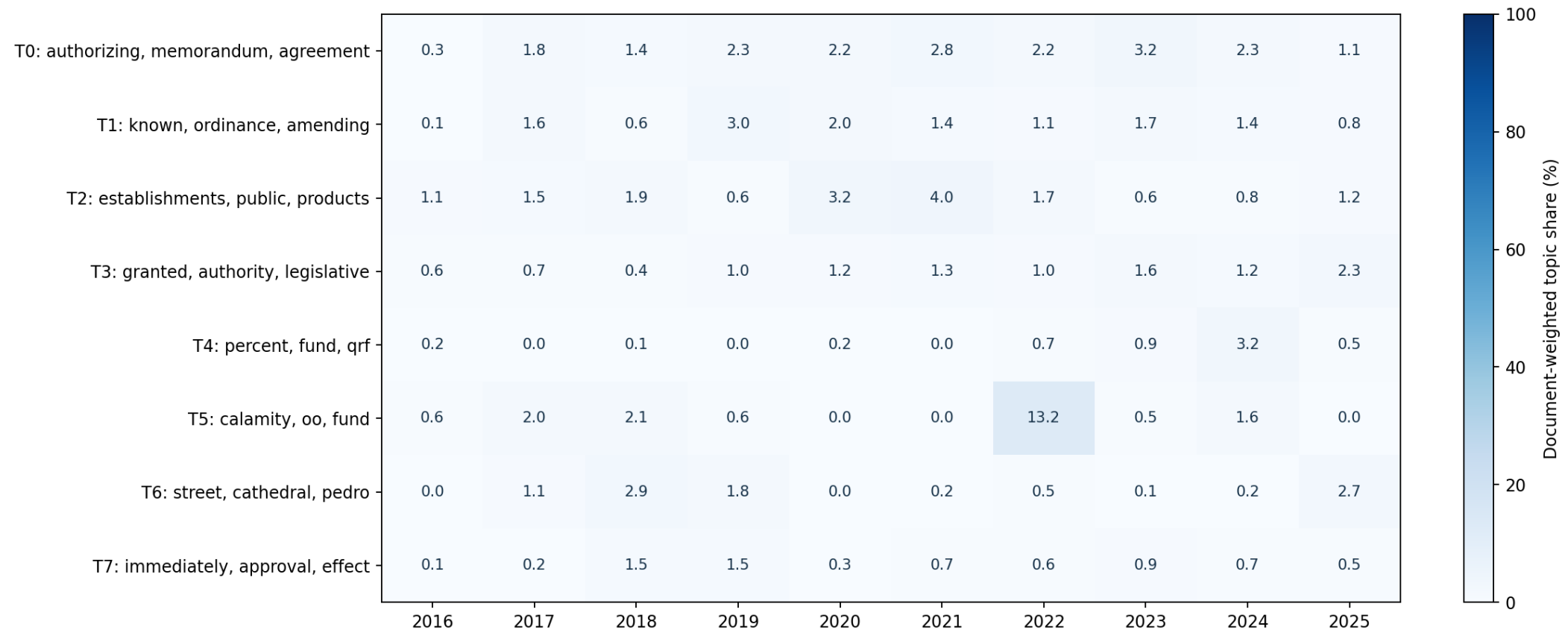
TITLE.- This Ordinance shall be known as..AN
ORDINANCE FOR THE TEMPORARY CLOSURE TO
VEHICULAR TRAFFIC OF THE ROAD BESIDE RIZAL PARK
ALONG BOLTON AND SAN PEDRO STREETS,
POBLACION, THIS CITY, ON NOVEMBER 30, 2016 FROM
1:00 P.M. TO 10:00 p.M. FOR A RELIGIOUS PUBLIC
RALLY UNDER THE THEME, "JESUS'S REIGNS

Yearly assignment coverage



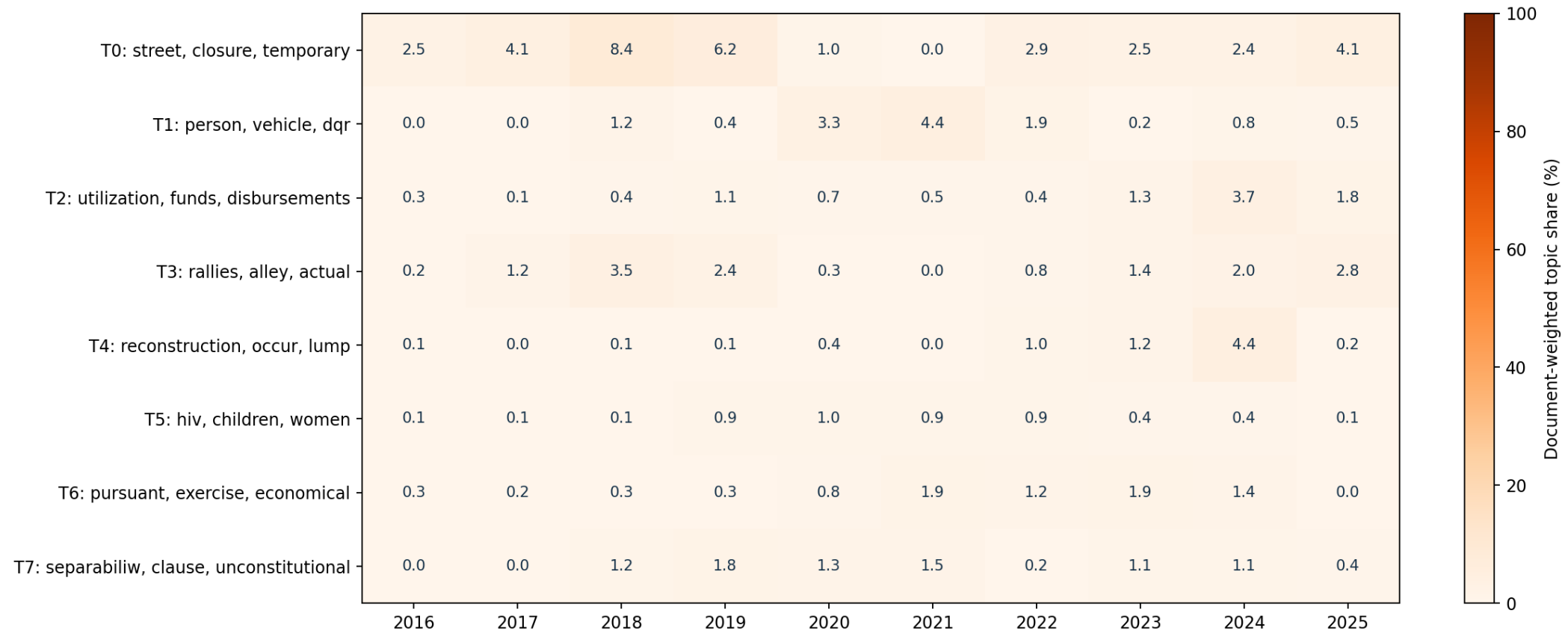
Each year's denominator includes all modeled units, including outliers; corpus sizes differ across years.

Model A · Lexical · document-weighted trends



Each source document contributes total weight 1, distributed across its unit topics. Top eight topics only; columns need not sum to 100%.

Model B · Legal-BERT · document-weighted trends



Each source document contributes total weight 1, distributed across its unit topics. Top eight topics only; columns need not sum to 100%.

Evaluation contract and remaining limitations

C_v: shared unigram keywords and the same model-input texts.

Diversity: unique top keywords / actual keyword slots, excluding outliers.

Coverage: assigned units / all units. Seed ARI includes outliers as one cluster.

Yearly unit shares and document-weighted shares answer different questions.

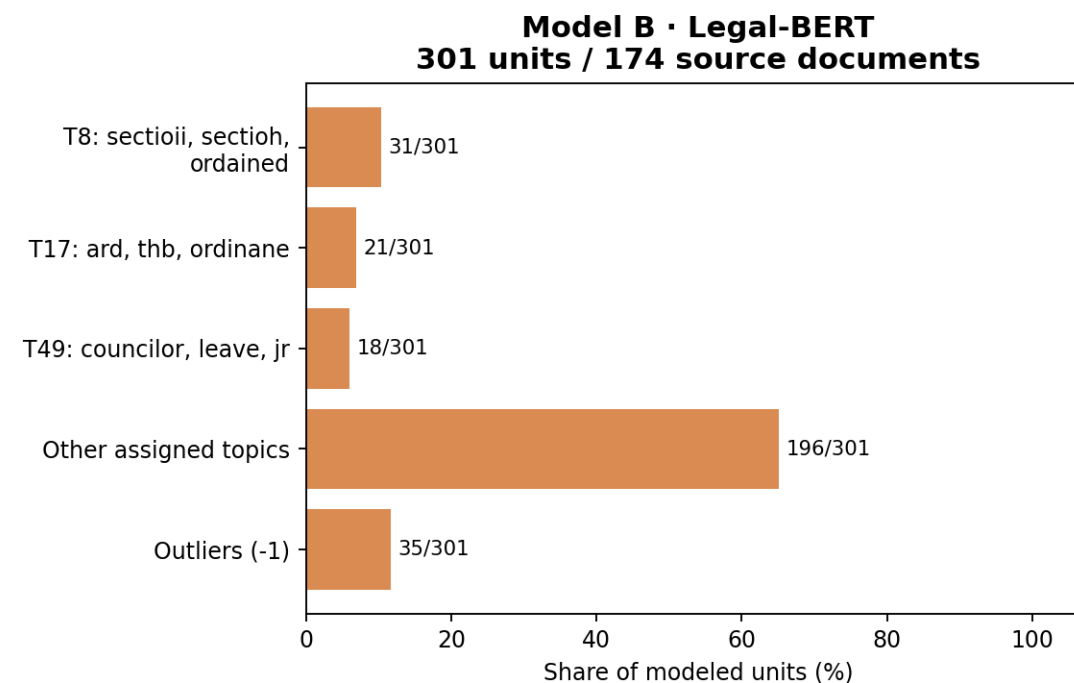
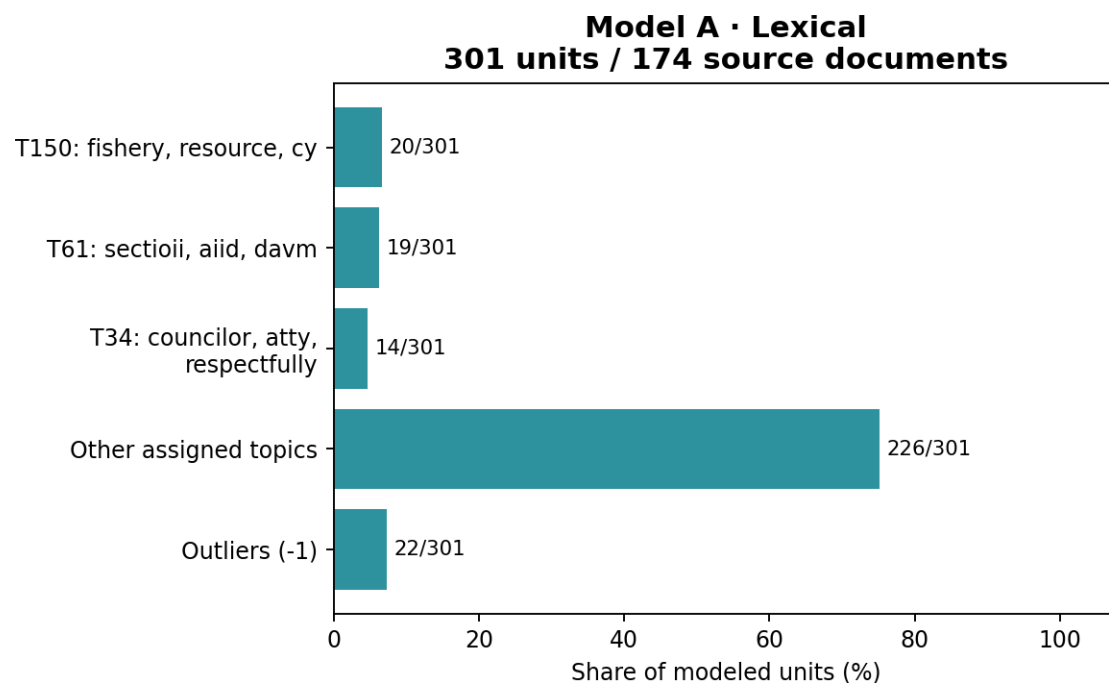
No artificial BERTopic perplexity, supervised accuracy or fabricated expert scores.

Remaining work: inspect extraction/date/version flags, review topic labels with experts, and predefine ordinance-grouped holdouts before final tuning.

Source verification does not itself validate clusters or temporal policy interpretations.

2016 · both models, full eligible texts

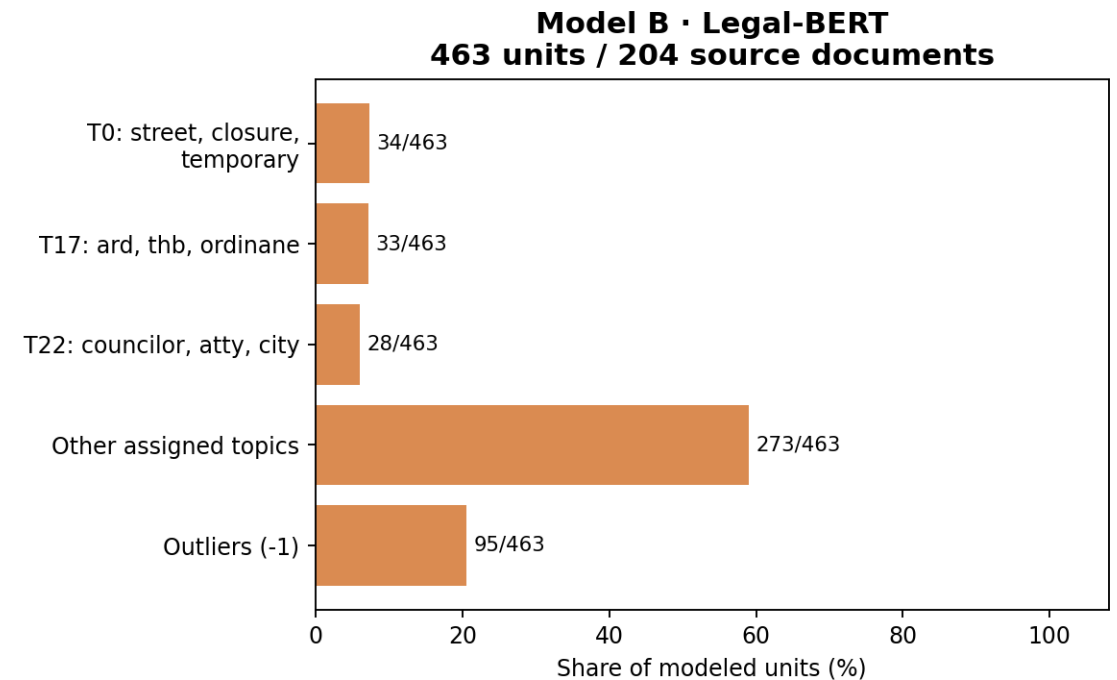
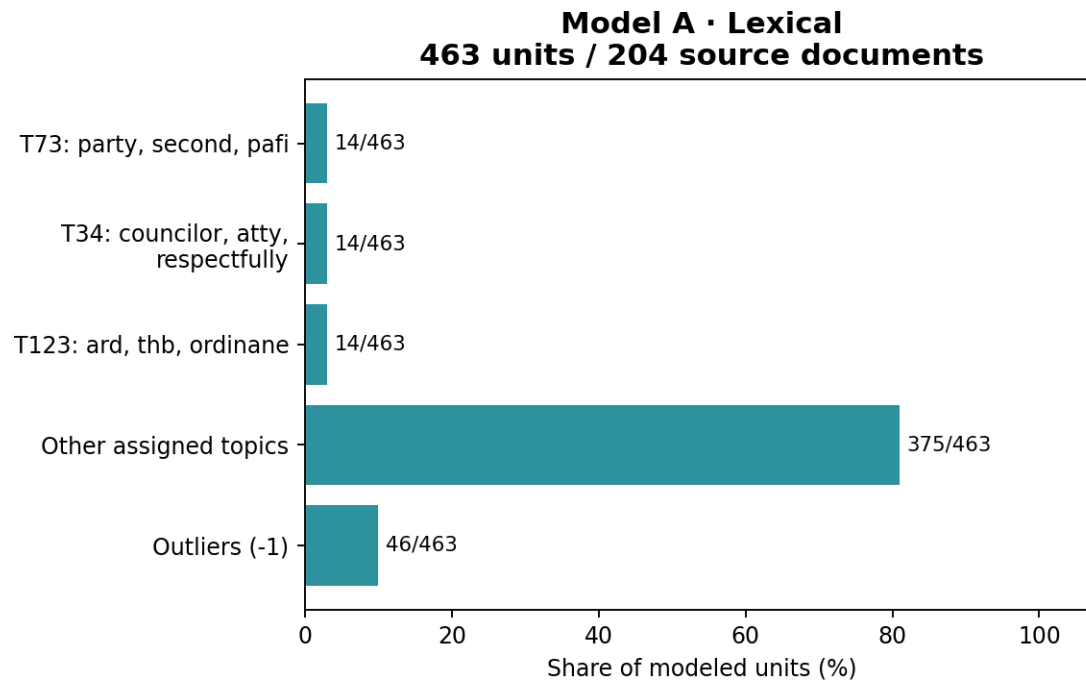
2016 · full eligible clean-text corpus



Topic numbers are model-local: A/T0 does not mean B/T0. Bars include top three topics, other assigned topics and outliers.

2017 · both models, full eligible texts

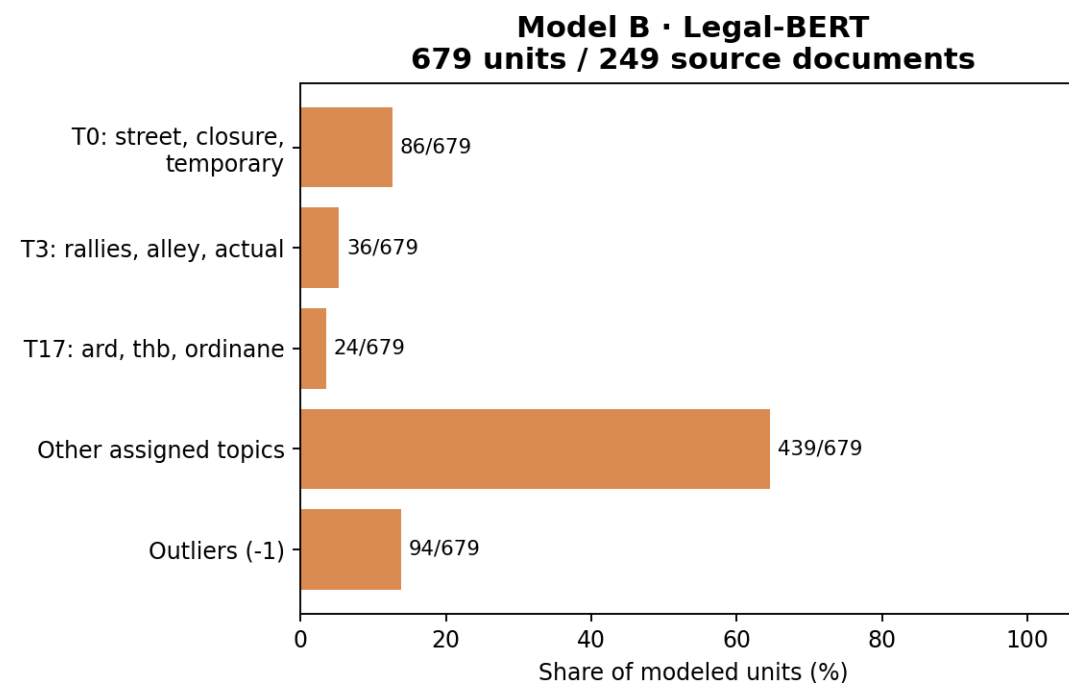
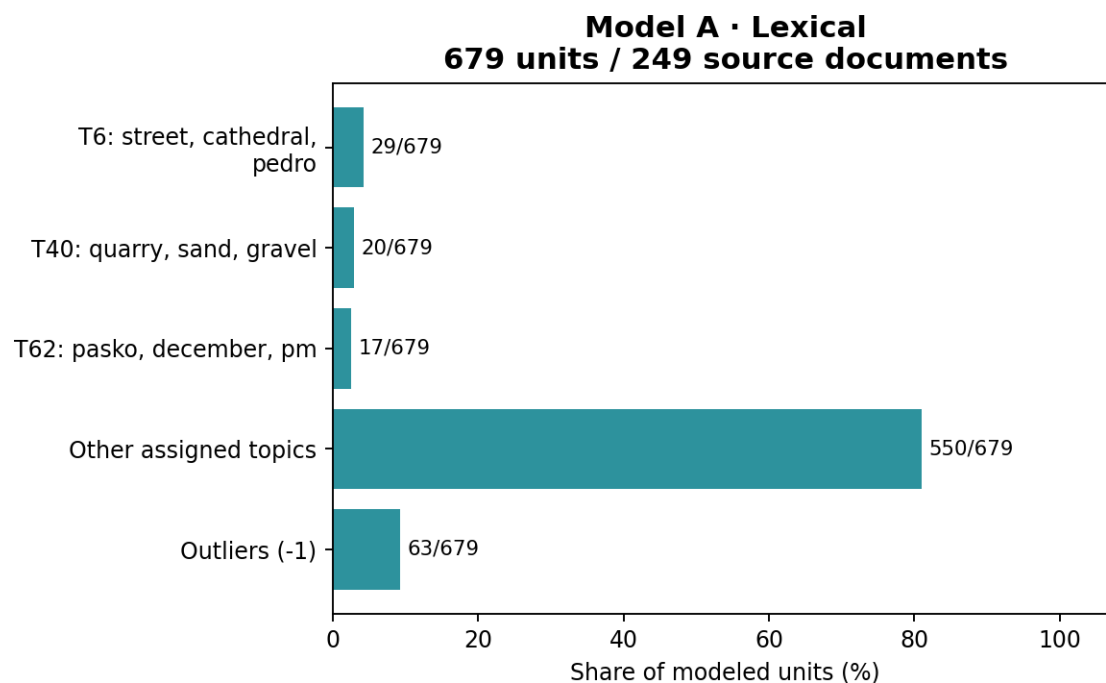
2017 · full eligible clean-text corpus



Topic numbers are model-local: A/T0 does not mean B/T0. Bars include top three topics, other assigned topics and outliers.

2018 · both models, full eligible texts

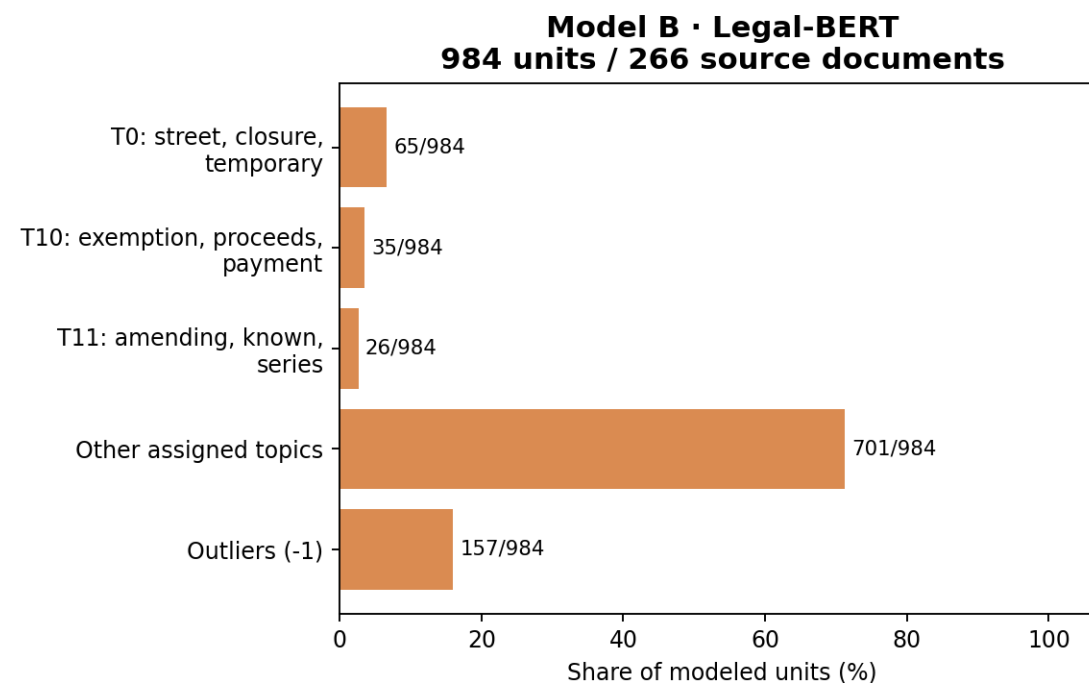
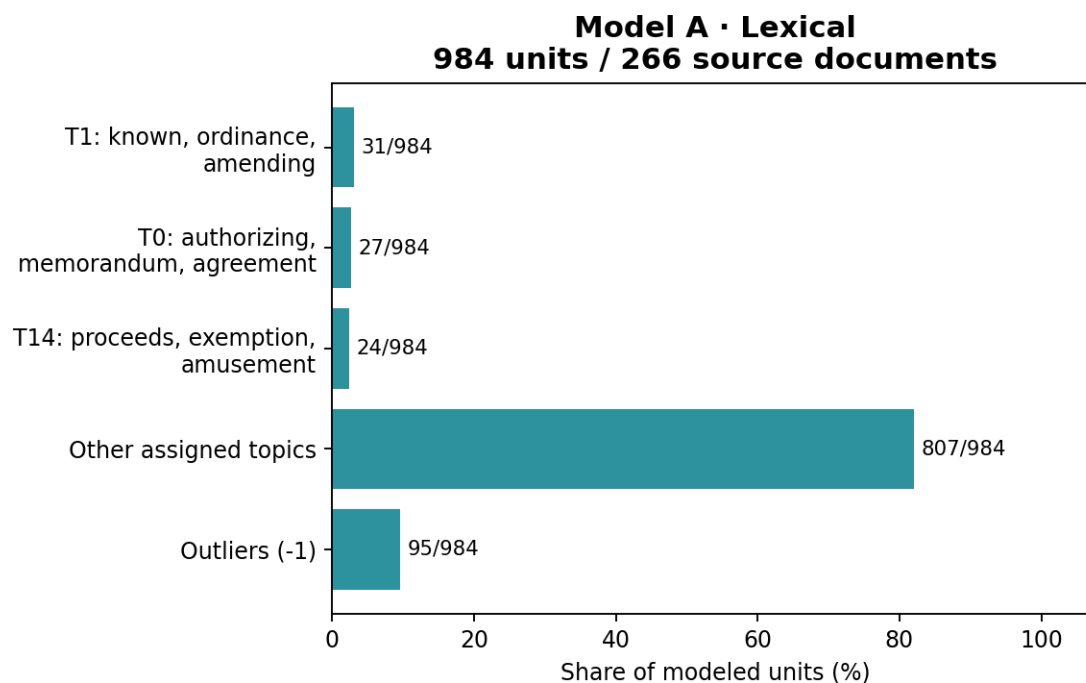
2018 · full eligible clean-text corpus



Topic numbers are model-local: A/T0 does not mean B/T0. Bars include top three topics, other assigned topics and outliers.

2019 · both models, full eligible texts

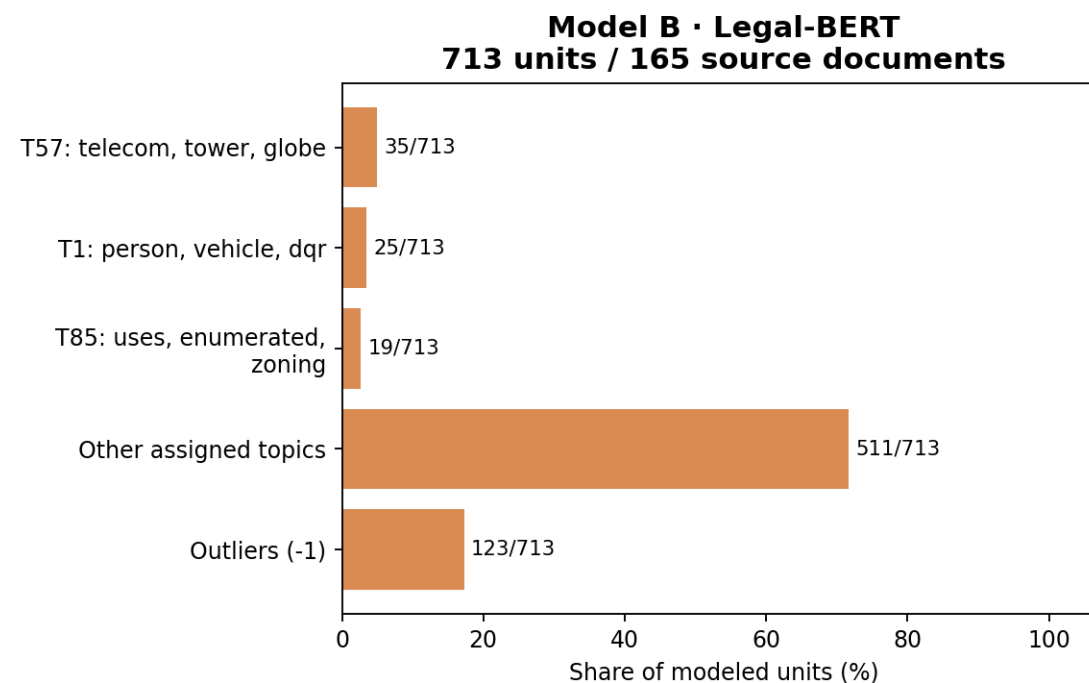
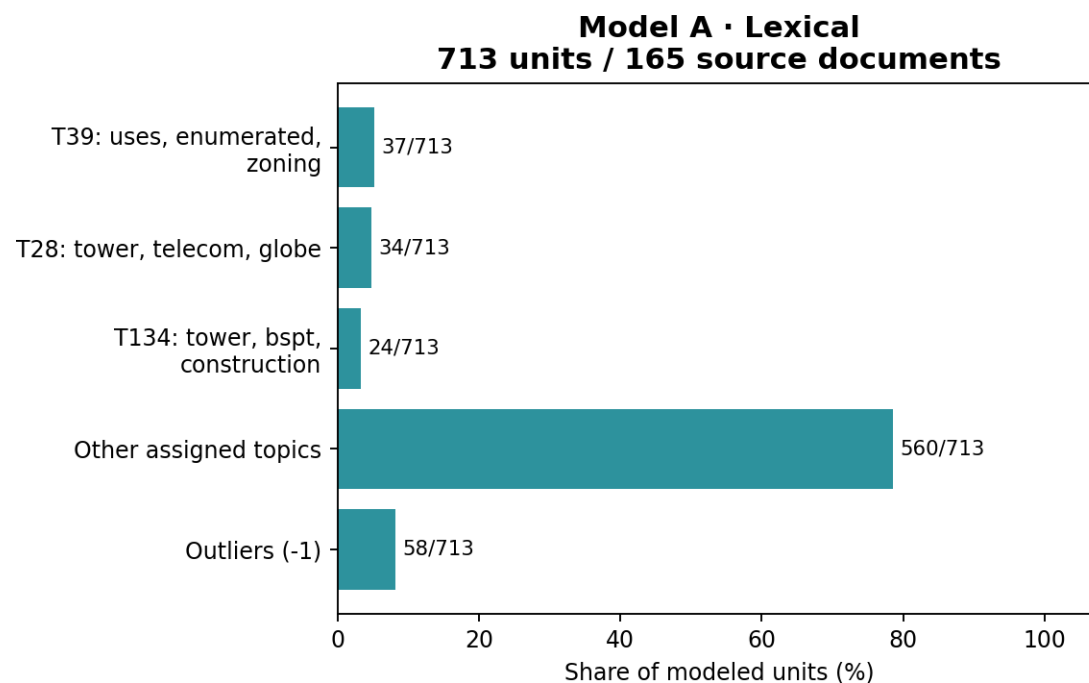
2019 · full eligible clean-text corpus



Topic numbers are model-local: A/T0 does not mean B/T0. Bars include top three topics, other assigned topics and outliers.

2020 · both models, full eligible texts

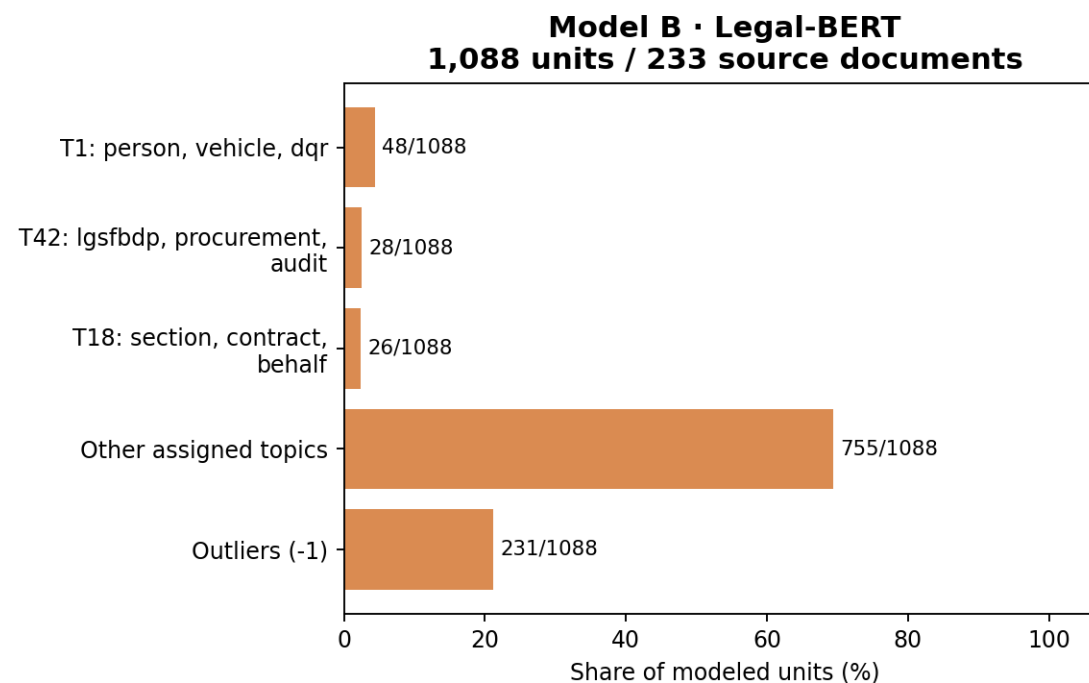
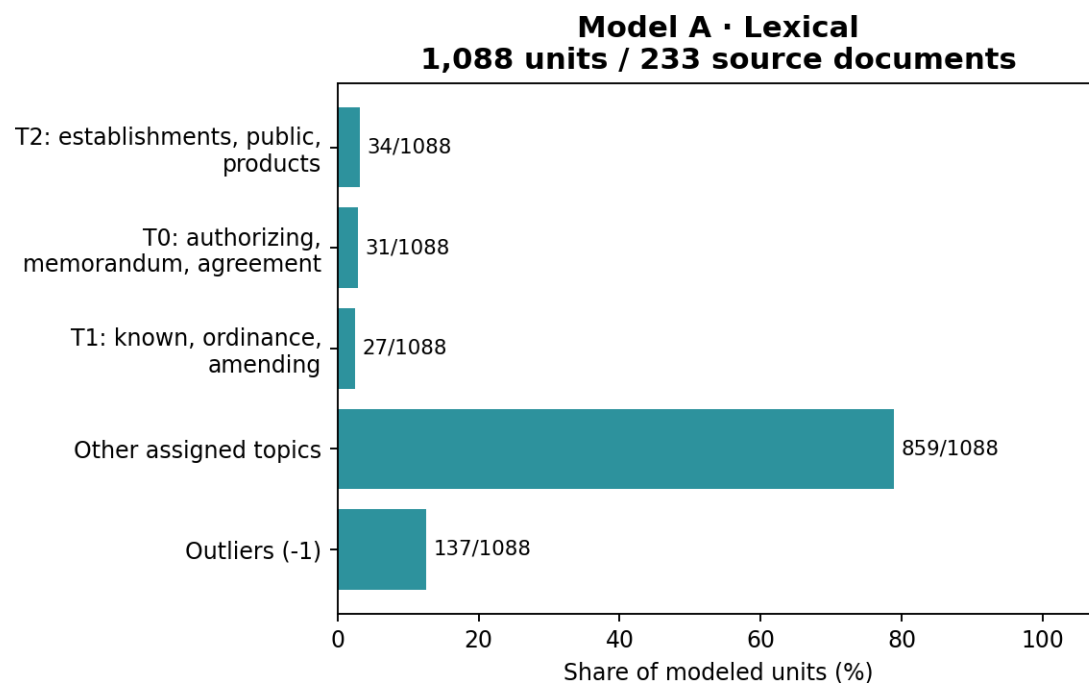
2020 · full eligible clean-text corpus



Topic numbers are model-local: A/T0 does not mean B/T0. Bars include top three topics, other assigned topics and outliers.

2021 · both models, full eligible texts

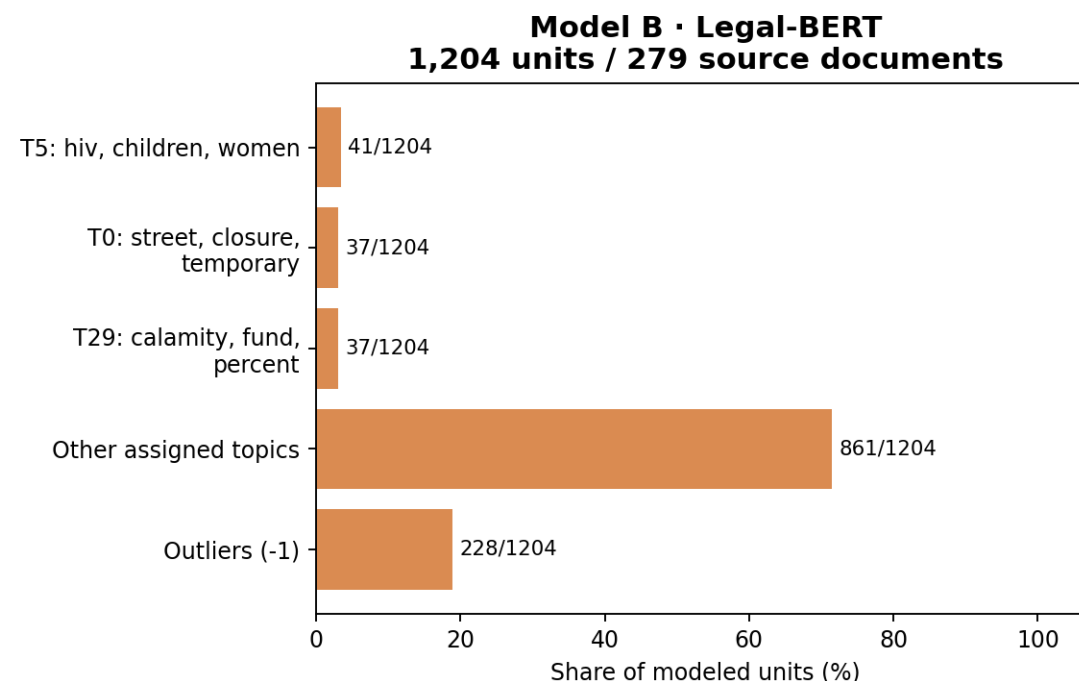
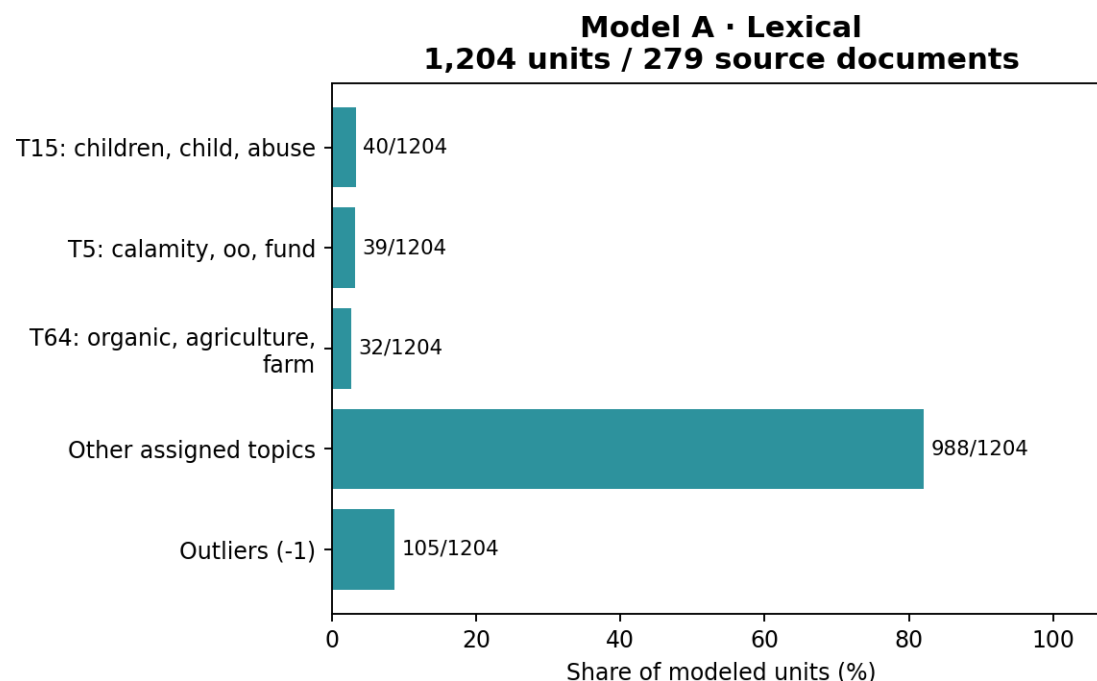
2021 · full eligible clean-text corpus



Topic numbers are model-local: A/T0 does not mean B/T0. Bars include top three topics, other assigned topics and outliers.

2022 · both models, full eligible texts

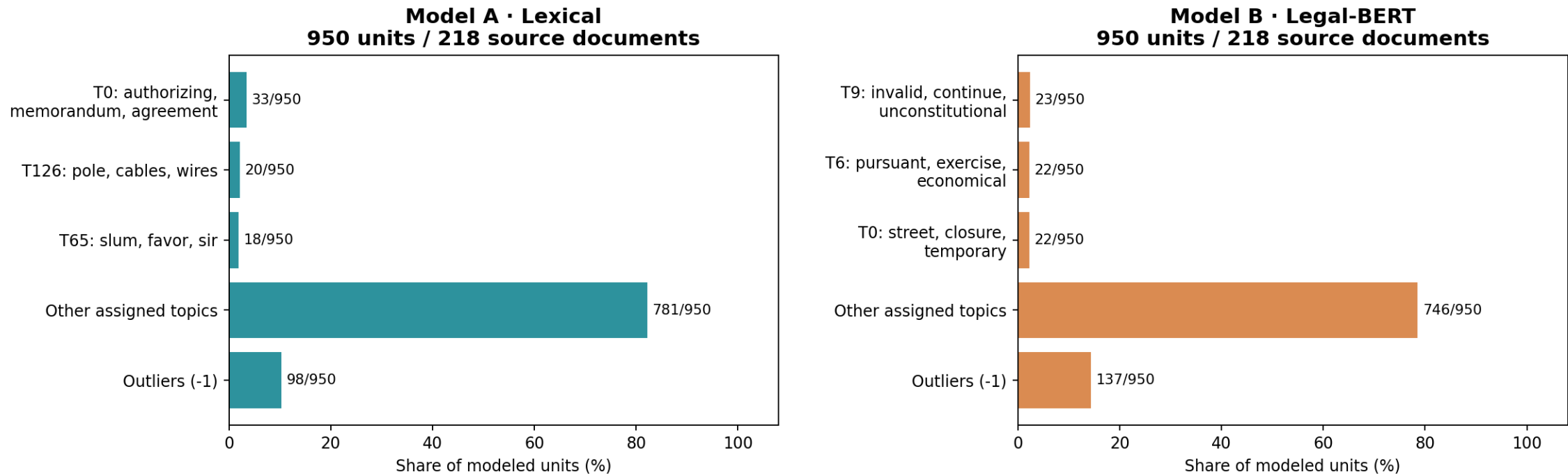
2022 · full eligible clean-text corpus



Topic numbers are model-local: A/T0 does not mean B/T0. Bars include top three topics, other assigned topics and outliers.

2023 · both models, full eligible texts

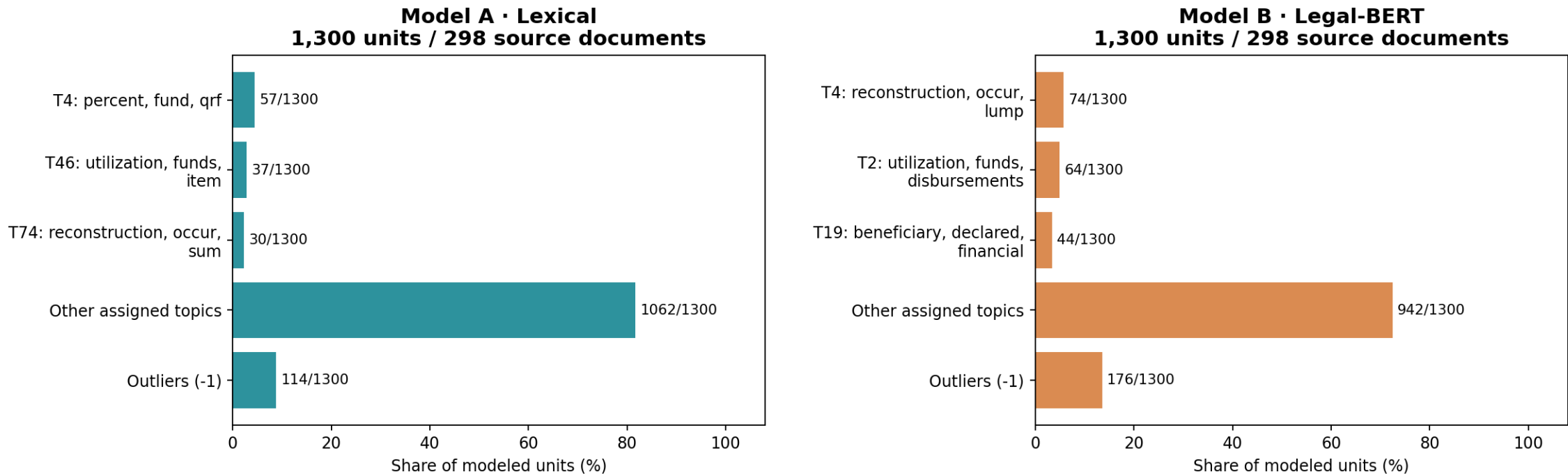
2023 · full eligible clean-text corpus



Topic numbers are model-local: A/T0 does not mean B/T0. Bars include top three topics, other assigned topics and outliers.

2024 · both models, full eligible texts

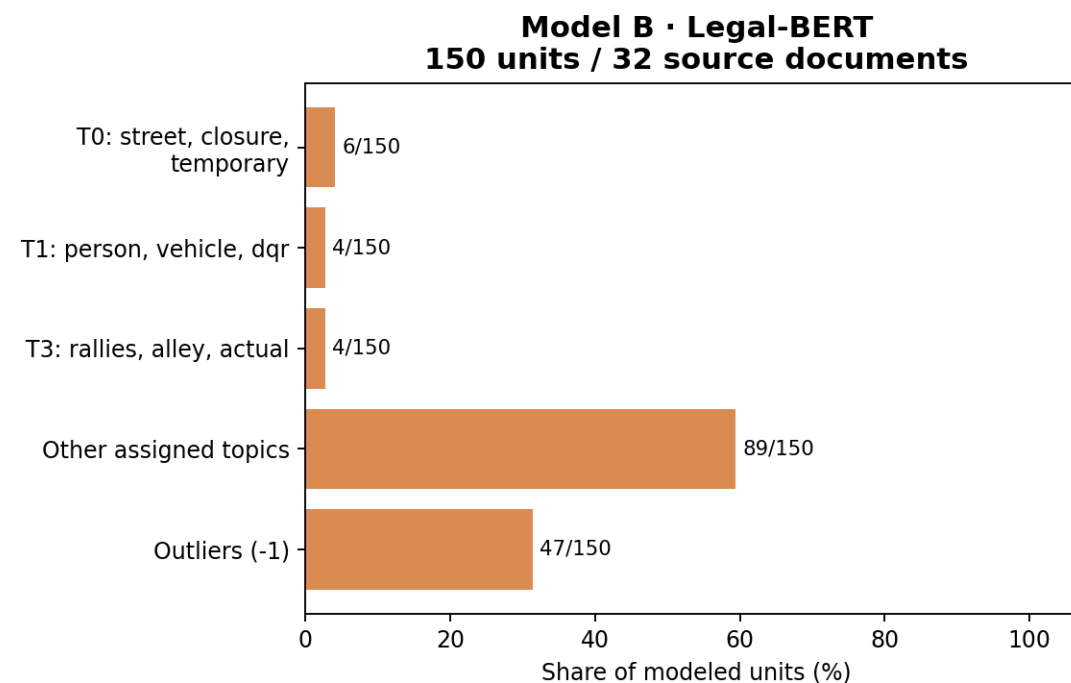
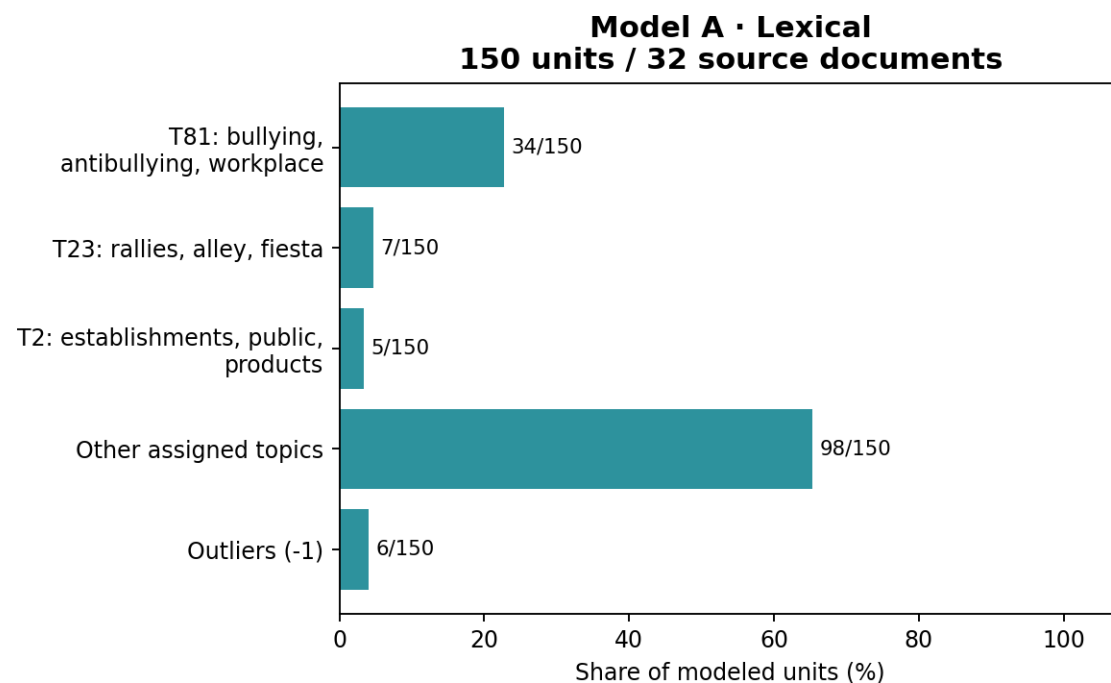
2024 · full eligible clean-text corpus



Topic numbers are model-local: A/T0 does not mean B/T0. Bars include top three topics, other assigned topics and outliers.

2025 · both models, full eligible texts

2025 · full eligible clean-text corpus



Topic numbers are model-local: A/T0 does not mean B/T0. Bars include top three topics, other assigned topics and outliers.